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POINT-OF-CARE DIAGNOSTIC SYSTEMS AND CONTAINERS FOR SAME

Abstract

The present disclosure relates to a medical diagnostic system. In various embodiments, the system includes a housing, a first receptacle in the housing for receiving a reagent container, a second receptacle in the housing for receiving a working fluid and waste container, where the second receptacle is larger than the first receptacle, two reagent access needles positioned and fixed within the first receptacle with each of the two reagent access needles being positioned to access the reagent container, and a working fluid access needle and a waste access needle positioned and fixed within the second receptacle with the working fluid access needle and the waste access needle being positioned to access the working fluid and waste container.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. **17**/**836**,474, now U.S. Pat. No. 12,303,902, which is a continuation of U.S. patent application Ser. No. 16/378,483, now U.S. Pat. No. 11,358,148, which is a continuation-in-part of U.S. patent application Ser. No. 15/941,596, now U.S. Pat. No. 11,541,396. The entire contents of each of the foregoing applications are hereby incorporated by reference herein.

TECHNICAL FIELD
[0002] The present disclosure relates to medical diagnostics, and more particularly, to point-of-care medical diagnostic systems.
BACKGROUND
[0003] Medical guidance for many medical diagnostic systems, such as hematology analyzers, recommends analyzing a sample as soon as possible after drawing the sample. This recommendation can be difficult if the sample is obtained at the point of care but the test is to be performed at an external laboratory. Therefore, many doctors and veterinarians prefer to have point-of-care (POC) systems to analyze fresh samples.
[0004] POC medical diagnostic systems use various types of reagents and fluids to perform their analyses. Various types of packages exist for the reagents and fluids, and such packages must be delivered to and installed by the POC offices. Installations requiring a multitude of steps can confuse and frustrate operators. In some cases, POC diagnostic systems may still operate even if packages are improperly installed but may produce incorrect results. Accordingly, there is continuing interest in improving POC medical diagnostic systems and reagent and fluid packages for POC medical diagnostic systems.
SUMMARY
[0005] The present disclosure relates to point-of-care medical diagnostic systems and containers for such systems.
[0006] In accordance with aspects of the present disclosure, a medical diagnostic system includes a housing, a first receptacle in the housing for receiving a reagent container, a second receptacle in the housing for receiving a working fluid and waste container where the second receptacle is larger than the first receptacle, two reagent access needles positioned and fixed within the first receptacle with each of the two reagent access needles being positioned to access the reagent container, and a working fluid access needle and a waste access needle positioned and fixed within the second receptacle with the working fluid access needle and the waste access needle being positioned to access the working fluid and waste container.
[0007] In various embodiments, the first receptacle includes a top wall, a bottom wall, side walls, and a back wall. One of the two reagent access needles is positioned on the back wall adjacent to the bottom wall. The other of the two reagent access needles is positioned on the back wall adjacent to and above a center line between the top and bottom walls. In various embodiments, the reagent

container and the first receptacle are shaped such that the reagent container must be inserted into the first receptacle in a particular orientation for the two reagent access needles to access the reagent container.

[0008] In various embodiments, the second receptacle includes a top wall, a bottom wall, side walls, and a back wall. The waste access needle is positioned on the back wall adjacent to the top wall. The working fluid access needle is positioned on the back wall adjacent to the bottom wall. In various embodiments, the working fluid and waste container and the second receptacle are shaped such that the working fluid and waste container must be inserted into the second receptacle in a particular orientation for the working fluid access needle and the waste access needle to access the working fluid and waste container. In various embodiments, a top portion of the second receptacle is narrower than a bottom portion of the second receptacle, and a top portion of the working fluid and waste container is narrower than a bottom portion of the working fluid and waste container.

[0009] In various embodiments, the medical diagnostic system includes a camera positioned such that it can view the first receptacle for imaging an encoded data-matrix code on the reagent container. In various embodiments, the medical diagnostic system uses the same camera positioned such that it can also view the second receptacle for imaging an encoded data-matrix code on the working fluid and waste container.

[0010] In various embodiments, the reagent container of the medical diagnostic system includes a top compartment and a bottom compartment that are fluidically separate. A septum between the top and bottom compartments connects them such that the top and bottom compartments are stationary relative to each other. The top compartment is defined by a housing having a top wall, a bottom wall, side walls, and an access opening positioned adjacent to the bottom wall of the top compartment. The bottom compartment is defined by a housing having a top wall, a bottom wall, side walls, and an access opening positioned adjacent to the bottom wall of the bottom compartment. In various embodiments, at least a portion of the bottom wall of the top compartment slopes downward toward the access opening of the top compartment. In various embodiments, the top wall of the bottom compartment is substantially parallel to the bottom wall of the top compartment. In various embodiments, a portion of the top wall of the bottom compartment is higher than a portion of the bottom wall of the top compartment.

[0011] In various embodiments, the working fluid and waste container of the medical diagnostic system includes a working fluid compartment having an access opening, a waste compartment having an access opening where the waste compartment is fluidically separate from the working fluid compartment, and a septum between and connecting the working fluid compartment and the waste compartment such that the working fluid compartment and the waste compartment are stationary relative to each other. In various embodiments, the second receptacle of the housing includes a top wall, a bottom wall, side walls, and a back wall. The access opening of the waste compartment is positioned adjacent to the top wall of the second receptacle, and the access opening of the working fluid compartment is positioned adjacent to the bottom wall of the second receptacle. In various embodiments, the waste compartment has an inner wall and an outer wall. The inner wall and the outer wall have a vertical cross-section in substantially a shape of a trapezoid with an open corner. The working fluid compartment has a first portion inward of the inner wall of the waste compartment and a second portion extending through the open corner where the second portion ends in the access opening of the working fluid compartment.

[0012] In accordance with aspects of the present disclosure, a container for a medical diagnostics system includes a top compartment defined by a housing having a top wall, a bottom wall, side walls, and an access opening positioned adjacent to the bottom wall of the top compartment, a bottom compartment defined by a housing having a top wall, a bottom wall, side walls, and an access opening positioned adjacent to the bottom wall of the bottom compartment, where the top compartment and the bottom compartment are fluidically separate, and a septum between and connecting the top and bottom compartments such that the top and bottom compartments are

stationary relative to each other.
[0013] In various embodiments, at least a portion of the bottom wall of the top compartment slopes downward toward the access opening of the top compartment. In various embodiments, the top wall of the bottom compartment is substantially parallel to the bottom wall of the top compartment.
[0014] In accordance with aspects of the present disclosure, a container for a medical diagnostics system includes a waste compartment having an access opening, an inner wall, and an outer wall, where the inner wall and the outer wall have a vertical cross-section in substantially a shape of a trapezoid with an open corner, a working fluid compartment having a first portion inward of the inner wall of the waste compartment and a second portion extending through the open corner, where the second portion ends in an access opening of the working fluid compartment and where the working fluid compartment is fluidically separate from the waste compartment, and a septum between and connecting the working fluid compartment and the waste compartment such that the working fluid compartment and the waste compartment are stationary relative to each other.
[0015] In various embodiments, the container is configured to fit into a receptacle having a top wall, a bottom wall, side walls, and a back wall, the access opening of the waste compartment is positioned adjacent to the top wall of the receptacle, and the access opening of the working fluid compartment is positioned adjacent to the bottom wall of the receptacle.
[0016] In accordance with aspects of the present disclosure, a medical diagnostic system includes a housing, a receptacle in the housing for receiving a container having a compartment where the receptacle includes a wall, and a compartment engagement mechanism for engaging the compartment. The compartment engagement mechanism includes a hub fixed within the wall of the receptacle, at least one needle secured to the hub, a needle cover slideably coupled with the hub and having a cover position that covers the needle, and at least one engagement arm secured to the hub and grasping the needle cover when the needle cover is in the cover position.
[0017] In various embodiments, the at least one needle includes a coated needle and at least one non-coated needle.
[0018] In various embodiments, the needle cover includes a cap having at least one aperture corresponding to the at least one needle, and a plurality of glide posts slideably coupled with the hub, wherein the at least one needle protrudes through the at least one aperture of the cap when the plurality of glide posts slides into the hub.
[0019] In various embodiments, the at least one engagement arm includes a base portion secured to the hub, an elbow portion connected to the base portion, and a grasping portion connected to the elbow portion, where the elbow portion is semi-flexible and permits the grasping portion to move radially closer to and away from the needle cover.
[0020] In various embodiments, the medical diagnostic system further includes the container, where the container includes a cap secured to a neck of the compartment, and the cap is sized to contact the grasping portion of the at least one engagement arm and move the grasping portion radially away from the needle cover when the compartment engagement mechanism engages the compartment of the container.
[0021] In various embodiments, the compartment includes a collar having at least one notch corresponding to the at least one engagement arm, where the at least one notch of the collar receives the grasping portion of at least one engagement arm when the compartment of the container is fully engaged with the compartment engagement mechanism.
[0022] In various embodiments, the container includes a second compartment having a second neck and a second cap secured to the second neck, and the medical diagnostic system further includes a second compartment engagement mechanism coupled to the wall of the receptacle for engaging the second compartment of the container.
[0023] In various embodiments, the medical diagnostic system further includes a level detection mechanism coupled to the compartment engagement mechanism and the second compartment engagement mechanism, where the level detection mechanism detects whether the container is

level within the receptacle.
[0024] In various embodiments, the compartment engagement mechanism is a different size from the second compartment engagement mechanism, and the cap is a different size from the second cap. In various embodiments, the compartment engagement mechanism and the second compartment engagement mechanism extend different distances from the wall of the receptacle, and the cap and the second cap extend different distances from a body of the container.
[0025] Further details and aspects of exemplary embodiments of the present disclosure are described in more detail below with reference to the appended figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0026] FIG. **1** is a diagram of an embodiment of a medical diagnostic system in accordance with aspects of the present disclosure;
[0027] FIG. **2** is a diagram of an embodiment of a reagent container, in accordance with aspects of the present disclosure;
[0028] FIG. **3** is a diagram of a cross-section of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0029] FIG. **4** is a diagram of another embodiment of a reagent container, in accordance with aspects of the present disclosure;
[0030] FIG. **5** is a diagram of an embodiment of a working fluid and waste container, in accordance with aspects of the present disclosure;
[0031] FIG. **6** is a diagram of a cross-section of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0032] FIG. **7** is a diagram of another embodiment of a working fluid and waste container, in accordance with aspects of the present disclosure;
[0033] FIG. **8** is a diagram of yet another embodiment of a working fluid and waste container, in accordance with aspects of the present disclosure;
[0034] FIG. **9** is a diagram of still another embodiment of a working fluid and waste container, in accordance with aspects of the present disclosure;
[0035] FIG. **10** is a diagram of another embodiment of a working fluid and waste container, in accordance with aspects of the present disclosure;
[0036] FIG. **11** is a perspective view of the reagent container of FIG. **2** with contour lines that more clearly show the shape of the container, in accordance with aspects of the present disclosure;
[0037] FIG. **12** is a bottom plan view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0038] FIG. **13** is a top plan view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0039] FIG. **14** is a left side elevational view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0040] FIG. **15** is a right side elevational view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0041] FIG. **16** is a front elevational view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0042] FIG. **17** is a rear elevational view of the reagent container of FIG. **2**, in accordance with aspects of the present disclosure;
[0043] FIG. **18** is a perspective view of the working fluid and waste container of FIG. **5** with contour lines that more clearly show the shape of the container, in accordance with aspects of the present disclosure;

[0044] FIG. **19** is a bottom plan view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0045] FIG. **20** is a top plan view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0046] FIG. **21** is a left side elevational view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0047] FIG. **22** is a right side elevational view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0048] FIG. **23** is a front elevational view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0049] FIG. **24** is a rear elevational view of the working fluid and waste container of FIG. **5**, in accordance with aspects of the present disclosure;
[0050] FIG. **25** is a perspective view of another embodiment of a reagent container with contour lines that more clearly show the shape of the container, in accordance with aspects of the present disclosure;
[0051] FIG. **26** is a perspective view of another embodiment of a working fluid and waste container with contour lines that more clearly show the shape of the container, in accordance with aspects of the present disclosure;
[0052] FIG. **27** is a diagram of an embodiment of a container engagement mechanism having two compartment engagement mechanisms, in accordance with aspects of the present disclosure;
[0053] FIG. **28** is a perspective view of the compartment engagement mechanism of FIG. **27**, in accordance with aspects of the present disclosure;
[0054] FIG. **29** is another perspective view of the compartment engagement mechanism of FIG. **27**, in accordance with aspects of the present disclosure;
[0055] FIG. **30** is a diagram of certain components of the compartment engagement mechanism of FIG. **27**, in accordance with aspects of the present disclosure;
[0056] FIG. **31** is a diagram of certain other components of the compartment engagement mechanism of FIG. **27**, in accordance with aspects of the present disclosure;
[0057] FIG. **32** is a diagram of the container engagement mechanism of FIG. **27** before engaging a container, in accordance with aspects of the present disclosure;
[0058] FIG. **33** is a diagram of the container engagement mechanism of FIG. **27** after engaging a container, in accordance with aspects of the present disclosure;
[0059] FIG. **34** is a perspective view of another embodiment of a compartment engagement mechanism, in accordance with aspects of the present disclosure;
[0060] FIG. **35** is another perspective view of the compartment engagement mechanism of FIG. **34**, in accordance with aspects of the present disclosure;
[0061] FIG. **36** is a diagram of certain components of the compartment engagement mechanism of FIG. **34**, in accordance with aspects of the present disclosure; and
[0062] FIG. **37** is another perspective of the components of FIG. **36**, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

[0063] The present disclosure relates to point-of-care medical diagnostic systems and containers for medical diagnostic systems. As used herein, point-of-care refers to a location where care is provided to human or animal patients, and a medical diagnostic system refers to a system that can analyze a sample obtained from a patient to diagnose a medical condition of the patient. Accordingly, a medical diagnostic system includes a patient sample analyzer, such as, but not limited to, a flow cytometer.
[0064] The following description will use flow-cytometry-based systems as an example of a medical diagnostic system. An example of a flow-cytometry-based analyzer is shown and described in U.S. Pat. No. 7,324,194, which is hereby incorporated by reference herein in its entirety, and

which persons skilled in the art will understand. The present disclosure, however, is intended to and should be understood to apply to other types of medical diagnostic systems as well.

[0065] Flow cytometry systems include sub-systems such as fluidics, optics, and electronics sub-systems. A fluidics sub-system arranges a sample into a stream of particles, such as a stream of cells. The optics sub-system examines each cell by directed a laser beam to each cell and detecting scattered light using photo-detectors. Light is scattered according to size, complexity, granularity, and diameter of the cells, which form a “fingerprint” of each cell type. The electronics sub-system can process the fingerprints to classify, count, and/or otherwise analyze the cells/particles in the sample stream.

[0066] The fluidics sub-system has many responsibilities. For example, the fluidics sub-system uses a working fluid in various ways, including transporting dilutions (blood or quality control materials) to a laser for cell counting and morphology and/or to a hemoglobin module for hemoglobin measurement, acting as a sheath to carry blood cells sequentially past the laser, cleaning and/or priming the diagnostic system, and/or carrying waste to a waste container. The working fluid material is typically water-based and contains salt, surfactants, buffers and antimicrobials. The fluidic system is generally filled with this fluid at all times, except when a blood sample is being processed and moved through the system.

[0067] The fluidics sub-system also accesses reagents and applies them to the patient sample to produce desired reactions. For example, as persons skilled in the art will understand, reagents can be used to dye and distinguish particular cells, lyse red blood cells, and prepare cells for particular types of assays, among other things. In various embodiments, a red reagent is used to prepare a whole blood sample for evaluation primarily of red blood cells and platelets. The material is water-based and contains salt, surfactants, antimicrobials, and a stain (for reticulocytes). The red reagent is mixed in the proper dilution concentration with whole blood to cause the red blood cells to sphere and to stain the reticulocytes. The diluted sample is then transported to the flow cell for evaluation (counting and classification). In various embodiments, a white reagent is used to prepare a whole blood sample for evaluation of white blood cells. The material is water-based and contains salt, surfactants, and antimicrobials. The white reagent is mixed in the proper dilution concentration with whole blood to cause the red blood cells to lyse. The remaining white blood cells and platelets are left in the dilution and are transported to the flow cell for evaluation (counting and classification).

[0068] Accordingly, working fluid and reagents need to be installed and provided to the medical diagnostic system. Then, when the analysis is completed, waste fluids generated by the system need to be gathered and disposed in a safe manner. The following describe a medical diagnostic system and containers that address these concerns.

[0069] Referring now to FIG. **1**, there is shown an exemplary medical diagnostic system **100**. The illustrated medical diagnostic system **100** is configured and sized to reside within a point-of-care (POC) office. The illustrated system includes a housing **110** that forms the overall structure of the medical diagnostic system. The housing **110** includes a smaller receptacle **120** that is intended to receive a reagent container **122** and a larger receptacle **130** that is intended to receive a working fluid and waste container **132**. The reagent container **122** stores reagents that will be used by the diagnostic system **100**, and the working fluid and waste container **132** operates to provide working fluid to the diagnostic system **100** and to receive waste fluid from the diagnostic system **100**. At the right side of the housing **110**, another receptacle **140** can receive various fluids and materials, including a patient sample, system cleaning fluid, and quality control materials, among other things.

[0070] As will be described in more detail below, the receptacles **120**, **130** and the containers **122**, **132** are configured so that an operator can slide a container **122**, **132** into a receptacle. In accordance with one aspect of the present disclosure, the interiors of the receptacles **120**, **130** include fluid access needles (not shown). As the containers **122**, **132** slide into the receptacles **120**,

130, the needles engage access openings in the containers. In various embodiments, one or both of the receptacles **120**, **130** is horizontal or substantially horizontal such that an operator can slide a container **122**, **132** horizontally into a receptacle, and the fluid access needles (not shown) can also be oriented horizontally or have a substantially horizontal orientation. As used herein, the term “horizontal” refers to an orientation that is parallel to a surface or plane on which the diagnostic system **100** is placed. In various embodiments, the receptacles and access needles are substantially horizontal in that they are intended to be horizontal but may not be fully horizontal due to, for example, slight manufacturing imperfections or limitations, or slight loosening of the needles within the receptacle over time due to wear, or other material, manufacturing, or environmental imperfections. In various embodiments, one or both of the receptacles **120**, **130** are angled such that an operator can slide a container **122**, **132** into a receptacle non-horizontally, such as at a downward angle, and the fluid access needles (not shown) can also be oriented at an angle. In various embodiments, one or more fluid access needles can be oriented parallel to the direction at which a container **122**, **132** slides into a receptacle **120**, **130**. In various embodiments, one or more fluid access needles (not shown) can be oriented at an angle with respect to the direction at which a container **122**, **132** slides into a receptacle **120**, **130**. Combinations of the disclosed embodiments are contemplated, such that each receptacle, container, or access needle can be implemented by a different disclosed embodiment.

[0071] Referring now to FIG. **2**, there is shown a side view of an exemplary reagent container **200**. The reagent container includes a top compartment **210** and a bottom compartment **220**. The two compartments **210**, **220** are fluidically separate. A septum **230** between the top and bottom compartments **210**, **220** connects the two compartments and holds them stationary relative to each other. In various embodiments, the reagent container is a single molded vessel, where the two compartments and the septum between the two compartments are formed in the same molding process. The top compartment **210** and the bottom compartment **220** both end with an access opening **212**, **222**. The access openings **212**, **222** are positioned so that the reagent access needles **124**, **126** located within the smaller receptacle **120** of the diagnostic system can access them. In various embodiments, the access openings can be covered by a fluid seal that prevents the reagents from spilling. The reagent access needles **124**, **126** can puncture the fluid seal to access the reagents. The reagent access needles **124**, **126** are illustrated for clarity and are not a part of the reagent container.

[0072] FIG. **3** shows a vertical cross-section of the reagent container **200** of FIG. **2**. The top compartment **210** includes a top wall **214** and a bottom wall **216**, and the bottom compartment **220** includes a top wall **224** and a bottom wall **226**. Side walls of the top and bottom compartments **210**, **220** are illustrated in FIG. **2**. Depending on the location of the vertical cross-section, the area **302** between the top compartment **210** and the bottom compartment **220** may be the septum **230** or may be empty space. For example, as shown in FIG. **1**, the septum **230** is narrower than the widths of the top and bottom compartments **210**, **220**. If the vertical cross-section is taken over the septum **230**, then the septum **230** will be in the area **302** between the top and bottom compartments. If the vertical cross-section is taken outside of the septum **230**, then the area **302** between the top and bottom compartments **210**, **220** will be air. In various embodiments, the septum **230** can be wider or narrower or another width than as illustrated in FIG. **1**.

[0073] As shown in FIG. **3**, the access openings **212**, **222** of the top and bottom compartments **210**, **220** are adjacent to the bottom walls **216**, **226**. The reagent access needles **124**, **126** are positioned to insert into the bottom portion of the access openings **212**, **222**, so as to reach as much of the reagents as possible. A portion of the bottom wall **216** of the top compartment **210** slopes downward towards the access opening **212** of the top compartment. Thus, essentially all of the reagent in the top compartment **210** will be able to reach the access opening **212** and be accessed by the fluid access needle **124**. In contrast, in the illustrated embodiment, the bottom compartment **220** does not include a slope at its bottom wall **226**. Thus, some portion of the reagent in the bottom compartment **220** will be inaccessible to the fluid access needle **126**. In various embodiments, the bottom wall **226** of the bottom compartment **220** can include a downward slope.

[0074] The particular shapes and relative sizes of the compartments are exemplary, and other variations and configurations are contemplated. For example, in the embodiment of FIGS. **2** and **3**, the top compartment **210** is smaller than the bottom compartment **220**. For example, the top compartment **210** may hold from about 60 mL to about 130 mL, from about 70 mL to about 120 mL, from about 80 mL to about 110 mL, from about 90 mL to about 100 mL, or, most preferably, about 95 mL of reagent; and the bottom compartment **220** may hold from about 175 mL to about 100 mL, from about 165 mL to about 110 mL, from about 155 mL to about 120 mL, from about 145 mL to about 130 mL, or, most preferably, about 139 mL of reagent. In various other embodiments, other capacities are contemplated, and other proportions of capacities between the top and bottom compartments **210**, **220** are contemplated.

[0075] In the illustrated embodiment, the bottom wall **216** of the top compartment **210** and the top wall **224** of the bottom compartment **220** are parallel or substantially parallel. They may be substantially parallel even when they are intended to be entirely parallel because of, for example, manufacturing imperfections. In various other embodiments, the bottom wall **216** of the top compartment **210** and the top wall **224** of the bottom compartment **220** can be intentionally non-parallel. Additionally, in the illustrated embodiment, a portion of the top wall **224** of the bottom compartment **220** is higher than a portion of the bottom wall **216** of the top compartment **210** because of the downward slope in those walls. In various other embodiments, there may be no downward slope in those walls, such as in the example of FIG. **4**.

[0076] In the illustrated embodiment, the septum **230** adjacent to the access openings **212**, **222** is located about halfway between the top wall **214** of the top compartment **210** and the bottom wall **226** of the bottom compartment **220**. Thus, the access opening **212** of the top compartment **210** is located adjacent to and above this center line. The reagent access needles **124**, **126** are located in corresponding positions. The smaller receptacle **120** of the diagnostic system includes a top wall, a bottom wall, a back wall, and side walls (not shown). One reagent access needle **126** is positioned on the back wall adjacent to the bottom wall of the smaller receptacle **120**, and the other reagent access needle **124** is positioned on the back wall adjacent to and above the center line between the top and bottom walls of the smaller receptacle **120** (not shown). Thus, the reagent access needles **124**, **126** can access the compartments **210**, **220** only when the reagent container **200** is inserted into the smaller receptacle **120** in a particular orientation. In various other embodiments, the locations of the access openings **212**, **222** and the reagent access needles **124**, **126** can be in other positions, as shown for example, in FIG. **4**.

[0077] Described above herein are aspects of the medical diagnostic system and the reagent container. The following will describe aspects of the working fluid and waste container. As shown in FIG. **1**, the working fluid and waste container **132** is larger than the reagent container **122**. In various embodiments, other size proportions between the reagent container **122** and the working fluid and waste container **132** are contemplated.

[0078] Referring to FIG. **5**, there is shown an embodiment of a working fluid and waste container **500** that includes a working fluid compartment **510** and a waste compartment **520**. The working fluid and waste compartments **510**, **520** are fluidically separate. A septum **530** is positioned between the compartments and connects them such that the working fluid and waste compartments **510**, **520** are stationary relative to each other. In various embodiments, the working fluid and waste container is a single molded vessel, where the two compartments and the septum between the two compartments are formed in the same molding process. Referring also to FIG. **6**, a vertical cross-section of the working fluid and waste container **500** of FIG. **5** is shown. Depending on the location of the vertical cross-section, the area **602** between the waste compartment **520** and the working fluid compartment **510** may be the septum **530** or may be empty space. For example, as shown in FIG. **5**, the septum **530** is narrower than the widths of the working fluid and waste compartments

510, **520**. If the vertical cross-section is taken over the septum **530**, then the septum **530** will be in the area **602** between the working fluid and waste compartments **510**, **520**. If the vertical cross-section is taken outside of the septum **530**, then the area **602** between the working fluid and waste compartments **510**, **520** will be air. In various embodiments, the septum **530** can be wider or narrower or another width than as illustrated in FIG. **5**.

[0079] With continuing reference to FIG. **6**, the waste compartment **520** has an outer wall **522**, an inner wall **524**, and end walls **526** connecting the outer and inner walls **522**, **524**. The outer wall **522** and the inner wall **524** have vertical cross-sections that are substantially in the shape of a square or rectangle with an open corner. In various embodiments, the vertical cross-sections may have another shape or substantially another shape, such as a trapezoid. The cross-sections may have substantially a particular shape, but not exactly a particular shape, because of, for example, rounded corners or manufacturing imperfections or material stress over time. In various embodiments, the cross-sectional shape can be oriented in different directions. For example, when the cross-sectional shape is a trapezoid, the base of the trapezoid can be oriented towards any side of the waste compartment **520**. The working fluid compartment **510** includes a portion that is within the inner wall **524** of the waste compartment **520** and another portion **512** that extends through the open corner of the waste compartment **520** and ends at an access opening **512**.

[0080] With reference to the medical diagnostic system of FIG. **1**, the working fluid and waste container **132** slides into the larger receptacle **130**. The larger receptacle **130** includes a top wall, a bottom wall, a back wall, and side walls (not shown). With reference to the larger receptacle **130**, the access opening **528** of the waste compartment **520** is positioned adjacent to the top wall of the larger receptacle **130**, and the access opening **514** of the working fluid compartment **510** is positioned adjacent to the bottom wall of the second receptacle **130**. The working fluid and waste access needles **134**, **136** are located in corresponding positions. The waste access needle **134** is positioned on the back wall of the larger receptacle **130** adjacent to the top wall of the larger receptacle **130**, and the working fluid access needle **136** is positioned on the back wall of the larger receptacle **130** adjacent to the bottom wall of the larger receptacle **130**. In various embodiments, the working fluid access needle **136** is positioned towards the bottom portion of the access opening **514** for the working fluid compartment **510**. In this manner, substantially all of the working fluid is accessible to the working fluid access needle **136**. In various embodiments, the waste access needle **134** is positioned towards the top portion of the access opening **528** for the waste compartment **520**. In this manner, the waste compartment **520** can be filled without the stale waste fluid in the waste compartment **520** contacting the waste access needle **134**, thereby providing less risk of contaminating the waste access needle **134** or of backflow through the waste access needle **134**. In various embodiments, the access openings **514**, **528** can be covered by a fluid seal that prevents the fluid from spilling. The working fluid and waste access needles **134**, **136** can puncture the fluid seal to access the interior of the compartments **510**, **520**.

[0081] Referring again to FIG. **1**, in accordance with aspects of the present disclosure, the working fluid and waste container **132** and the larger receptacle **130** are shaped such that the working fluid and waste container **132** must be inserted into the larger receptacle **130** in a particular orientation for the working fluid access needle **136** and the waste access needle **134** to access the working fluid and waste container **132**. In various embodiments, the top portion of the larger receptacle **130** is narrower than the bottom portion of the larger receptacle **130**, and the top portion of the working fluid and waste container **132** is also narrower than the bottom portion of the working fluid and waste container **132**. Thus, the working fluid and waste container **132** must be inserted in the correct orientation for the working fluid access needle **136** and the waste access needle **134** to access the access openings of the working fluid and waste container **130**.

[0082] The working fluid and waste container of FIGS. **5** and **6** is exemplary, and other shapes and configurations are contemplated to be within the scope of the present disclosure. For example, other embodiments of the working fluid and waste container are shown in FIGS. **7-10**.

[0083] In the embodiment of FIG. **7**, the working fluid compartment **710** has a substantially U-shape. The waste compartment **720** inter-locks with the working fluid compartment and has a portion above the working fluid compartment, a portion within the U-shape of the working fluid compartment, and a portion outside of and adjacent to the working fluid compartment. The working fluid compartment **710** and the waste compartment **720** are fluidically separate, and a septum connects between the compartment such that the working fluid compartment **710** and the waste compartment **720** are stationary relative to each other. The access opening **722** of the waste compartment **720** is adjacent to the top of the container, and the access opening **712** of the working fluid compartment **710** is adjacent to the bottom of the container.
[0084] In the embodiment of FIG. **8**, the working fluid and waste container includes a handle **802** at the top of the container. The handle **802** enables an operator to more easily carry the container when it is outside the medical diagnostic system. A handle **802** as shown in FIG. **8** can be applied to any other embodiment disclosed herein or any embodiment contemplated to be within the scope of the present disclosure. With continuing reference to FIG. **8**, the working fluid compartment **810** is substantially in the shape of a trapezoid, and the waste compartment **820** has a complementary shape such that the overall shape of working fluid and waste container is square or rectangular, when not considering the shape of the handle **802**. The working fluid compartment **810** and the waste compartment **820** are fluidically separate, and a septum connects between the compartment such that the working fluid compartment **810** and the waste compartment **820** are stationary relative to each other. The access opening **822** of the waste compartment **820** is adjacent to the top of the container, and the access opening **812** of the working fluid compartment **810** is adjacent to the bottom of the container.
[0085] In the embodiment of FIG. **9**, the working fluid and waste container includes a grip-enhancement **902**. The grip-enhancement **902** enables an operator to more easily handle the container when inserting the container into the medical diagnostics system or removing the container. A grip enhancement **902** as shown in FIG. **9** can be applied to any other embodiment disclosed herein or any embodiment contemplated to be within the scope of the present disclosure. With continuing reference to FIG. **9**, the working fluid compartment **910** is substantially in the shape of a square, and the waste compartment **920** has a complementary shape such that the overall shape of working fluid and waste container is square or rectangular. The working fluid compartment **910** and the waste compartment **920** are fluidically separate, and a septum connects between the compartments such that the working fluid compartment **910** and the waste compartment **920** are stationary relative to each other. The access opening **922** of the waste compartment **920** is adjacent to the top of the container, and the access opening **912** of the working fluid compartment **910** is adjacent to the bottom of the container.
[0086] In the embodiment of FIG. **10**, the working fluid and waste container includes a handle **1002** at a corner of the container. A handle **1002** as shown in FIG. **10** can be applied to any other embodiment disclosed herein or any embodiment contemplated to be within the scope of the present disclosure. With continuing reference to FIG. **10**, the working fluid compartment **1010** has substantially an L-shape, and the waste compartment **1020** has a substantially rectangular shape, when not considering the shape of the handle **1002**. The working fluid compartment **1010** and the waste compartment **1020** are fluidically separate, and a septum connects between the compartments such that the working fluid compartment **1010** and the waste compartment **1020** are stationary relative to each other. The access opening **1022** of the waste compartment **1020** is adjacent to the top of the container, and the access opening **1012** of the working fluid compartment **1010** is adjacent to the bottom of the container.
[0087] Accordingly, describe above are a medical diagnostic system and containers for the medical diagnostic system. FIGS. **11-17** show additional views of the reagent container of FIG. **2**, with contour lines that more clearly illustrate the shape of the reagent container. In particular, FIG. **11** is a perspective view of the reagent container, FIG. **12** is a bottom plan view thereof, FIG. **13** is a top

plan view thereof, FIG. **14** is a left side elevational view thereof, FIG. **15** is a right side elevational view thereof, FIG. **16** is a front elevational view thereof, and FIG.

[0088] **17** is a rear elevational view thereof. As shown in FIGS. **11-17**, the perimeter of the illustrated reagent container is wider than the rest of the reagent container and permits an operator to more readily grasp the reagent container along any portion of the perimeter. FIGS. **18-24** show additional views of the working fluid and waste container of FIG. **5**, with contour lines that more clearly illustrate the shape of the working fluid and waste container. In particular, FIG. **18** is a perspective view of the working fluid and waste container, FIG. **19** is a bottom plan view thereof, FIG. **20** is a top plan view thereof, FIG. **21** is a left side elevational view thereof, FIG. **22** is a right side elevational view thereof, FIG. **23** is a front elevational view thereof, and FIG. **24** is a rear elevational view thereof. As shown in FIGS. **18-24**, the perimeter of the illustrated working fluid and waste container is wider than the rest of the container and permits an operator to more readily grasp the working fluid and waste container along any portion of the perimeter. The embodiments disclosed herein are merely exemplary and are not intended to limit the scope of the present disclosure.

[0089] The following describes a feature of present disclosure with reference to FIG. **1**. In accordance with aspects of the present disclosure, the medical diagnostic system includes one or more cameras (not shown) for imaging encoded data-matrix codes on the reagent container **122** and the working fluid and waste container **132**. The data-matrix codes on the containers **122**, **132** can encode information such as expiration date, lot number, manufacturer identity, and authenticity, among other things. The camera can scan the data-matrix code to read this information, and the medical diagnostic system **100** can process and respond to the information in various ways.

[0090] In various embodiments, the larger receptacle **130** uses the camera (not shown) for imaging a data-matrix code on the working fluid and waste container **132**, and the smaller receptacle **120** uses the same camera (not shown) for imaging a data-matrix code the reagent container **122**. In various embodiments, the data-matrix codes are positioned on the containers **122**, **132** such that the data-matrix codes can be read by the camera only when the containers **122**, **132** are inserted into the medical diagnostic system **100** in a particular orientation.

[0091] Accordingly, various aspects and embodiments of the present disclosure are described above. The following will describe other aspects and embodiments of the present disclosure.

[0092] Referring to FIG. **25**, there is shown an embodiment of a reagent container. Similar to the reagent container described in connection with FIG. **2** and FIG. **3**, the reagent container of FIG. **25** includes two fluidically separate compartments **2510** that end with access openings **2512**. The two compartments **2510** include reagents for use by the medical diagnostic system of FIG. **1**. Each access opening **2512** is at the end of a neck **2514** that includes threads for receiving a cap (not shown). The neck **2514** is situated in a collar **2516** of the reagent container. In various embodiments, the neck **2514** and the collar **2516** are formed integrally in the same manufacturing process. In various embodiments, the neck **2514** and the collar **2516** are formed separately in separate molding processes and are then secured to each other.

[0093] In accordance with aspects of the present disclosure, each collar **2516** includes one or more notches for receiving a container engagement mechanism, which will be described in connection with FIGS. **27-33**. In the illustrated embodiment, one notch **2518** can be seen for each collar **2516**. In various embodiments, the other side of the collar **2516** may or may not include another notch or other notches. In various embodiments, each collar **2516** can include multiple notches, such as two or more notches, which can be arranged or positioned at various regions of the collar **2516**. In various embodiments, different collars may have the same number of notches or different numbers of notches. In various embodiments, the notch or notches may have different shapes or dimensions than as illustrated.

[0094] FIG. **26** is shows an embodiment of a working fluid and waste container. Similar to the working fluid and waste container described in connection with FIG. **5** and FIG. **6**, the working fluid and waste container of FIG. **26** includes a working fluid compartment that is fluidically separate from a waste compartment. Each compartment **2610** ends in an access opening **2612**, and each access opening **2612** is at the end of a neck **2614** that includes threads for receiving a cap (not shown). The neck **2614** is secured to a collar **2616**, which can have one or more notches **2618** for receiving a container engagement mechanism. The embodiments and variations of the notches described in connection with FIG. **25** also apply to the collar of FIG. **26**.

[0095] Referring now to FIG. **27**, there is shown a diagram of an embodiment of a container engagement mechanism **2700**. The illustrated container engagement mechanism **2700** includes two compartment engagement mechanisms **2720**. With reference also to FIG. **1**, the container engagement mechanism **2700** can be positioned within the receptacles **120**, **130** of the medical diagnostic system **100** described in connection with FIG. **1**, such as positioned at the back walls **2702** of the receptacles **120**, **130**.

[0096] In the illustrated embodiment, each compartment engagement mechanism **2720** includes a hub **2722**, a needle **2724**, two or more engagement arms **2726**, a spring **2727**, and a needle cover **2728**. The hub **2722** is secured to and through a wall **2702** of the receptacle, such as the back wall of the receptacle. The hub **2722** holds and secures the needle **2724**, the engagement arms **2726**, and the needle cover **2728**. In various embodiments, the needle **2724** can be held to be perpendicular to or substantially perpendicular to the wall **2702** of the receptacle. In various embodiments, the needle **2724** can be firmly held by the hub **2722** such that the needle **2724** does not dislodge from the hub **2722** when it accesses a compartment of a container.

[0097] In accordance with aspects of the present disclosure, the hub **2722** holds the needle cover **2728** and serves as a guide to permit the needle cover **2728** to slide through or along the hub **2722**. The structure of the hub **2722** and the needle cover **2728** will be described in more detail in connection with FIGS. **28-30**. For the purpose of FIG. **27**, it is noted that the needle cover **2728** can slide through or along the hub **2722** in a direction that is parallel to the direction of the needle **2724** and/or that is perpendicular to the wall **2702** of the receptacle. The spring **2727** is positioned between the needle cover **2728** and the hub **2722** in a manner that biases the needle cover **2728** in a covered position that covers the needle **2724**.

[0098] The hub **2722** also holds the engagement arms **2726** such that the engagement arms **2726** secure the needle cover **2728** when the needle cover **2728** is in the covered position. The engagement arms **2726** are secured to the hub **2722** in a manner such that the engagement arms **2726** do not move through the hub **2722**. However, at least a portion of each engagement arm **2726** is semi-flexible so that each engagement arm can flex radially away from the needle **2724** and the needle cover **2728**. In various embodiments, a compartment engagement mechanism **2720** can include one engagement arm **2726**. In various embodiments, a compartment engagement mechanism **2720** can include more than one engagement arm **2726**. In various embodiments, the engagement arm(s) **2726** can be arranged or positioned in various ways different from the arrangement illustrated in FIG. **27**.

[0099] FIG. **28** shows a perspective view of the compartment engagement mechanism **2720** of FIG. **27** without the wall of the receptacle, including the hub **2722**, the needle cover **2728**, two engagement arms **2726**, and three needles **2724**. Although the illustrated embodiment includes three needles **2724**, another number of needles can be used in various embodiments, such as fewer than three needles or more than three needles. Additionally, the needles **2724** are arranged in a line in FIG. **28**. In various embodiments, multiple needles can be arranged in another manner, such as in a triangle or in another shape or another arrangement. As shown in FIG. **28**, the hub **2722** includes guide slots **2802** which receive a portion of the needle cover **2728** and permit the portion of the needle cover **2728** to slide through or along the hub **2722**.

[0100] FIG. **29** shows another perspective view of the compartment engagement mechanism **2720**, including a container-facing end of the mechanism. As shown in FIG. **29**, the container-facing end of the needle cover **2728** includes apertures **2902**. The apertures **2902** are aligned with the needles

2724 such that the needles **2724** extend through the apertures **2902** when the needle cover **2728** slides through or along the hub **2722**. The hub **2722** also includes slots **2904** that receive and secure the engagement arms **2726**. In various embodiments, when the engagement arms **2726** are secured to the hub **2722**, the engagement arms **2726** will no longer slide through the hub **2722**. FIGS. **28** and **29** are exemplary, and variations of or other embodiments of a compartment engagement mechanism are contemplated to be within the scope of the present disclosure.

[0101] FIG. **30** is a diagram of certain components of the container engagement mechanism **2700** of FIGS. **27-29**, including an engagement arm **2726**, a needle cover **2728**, and a hub **2722**. In the illustrated embodiment, the hub **2722** is formed from two identical half-components which combine to form the hub **2722** illustrated in FIGS. **27-29**. In various embodiments, the hub **2722** may be formed as one unitary component. In various embodiments, the hub **2722** can be made from a plastic material.

[0102] In the illustrated embodiment, the needle cover **2728** includes a cap **3002** and glide posts **3004** attached to the cap **3002**. In various embodiments, the cap **3002** and the glide posts **3004** can have various shapes, including shapes different from those illustrated in FIG. **30**. In various embodiments, the number of glide posts **3004** can vary, including fewer than four glide posts or more than four glide posts. The hub **2722** can include a corresponding number of guide slots **3012** to receive the glide posts **3004** of the needle cover **2728**. In various embodiments, the needle cover **2728** can be made from a plastic material.

[0103] The engagement arm **2726** includes a base portion **3022**, an elbow portion **3024**, and a grasping portion **3026**. The base portion **3022** is configured to sit within the hub **2722** and remain secured to the hub **2722** by ledges that are positioned to abut the hub **2722**. The hub **2722** includes a slot **3014** that receives the base portion **3022** of the engagement arm **2726**. The elbow portion **3024** of the engagement arm **2726** is connected to the base portion **3022**, and the grasping portion **3026** of the engagement arm **2726** is connected to the elbow portion **3024**. The elbow portion **3024** is semi-flexible so as to flex and permit the grasping portion **3026** to move radially closer to or farther away from the needle cover of the compartment engagement assembly. The end of the grasping portion **3026** includes a wedge shape pointed towards the needle cover of the compartment engagement assembly. In various embodiments, the engagement arm **2726** can be made from a plastic material.

[0104] FIG. **31** is a diagram of certain other components of the compartment engagement mechanism **2720**, including a level detection mechanism **3102**, a coated needle **3104**, an uncoated needle **3106**, and a spring **2727**. The spring **2727**, as described above, can be positioned to bias the needle cover **2728** in a covered position that covers the needle(s). In various embodiments, the coated needle **3104** can be coated with a material that has lower friction and that allows the needle **3104** to access a compartment more smoothly. In various embodiments, one or more uncoated needle **3106** may accompany a coated needle to provide greater flow to or from a container. In various embodiments, the needles are sharp and strong enough to puncture a cap of a container.

[0105] The level detection mechanism **3102** includes a cable having a detection end and a container end. At the container end, the cable is split into a number of leads **3108** that corresponds to the number of access openings of the container. The containers described above herein include two access openings. Therefore, the illustrated level detection mechanism **3102** splits into two leads **3108** at the container end. Each lead **3108** is connected to a needle cover. When both needle covers move at the same speed, tension remains on both leads **3108** and the detection end of the cable remains level. When one needle cover moves faster than the other needle cover, one lead loses tension and the detection end of the cable becomes non-level. Thus, if a container is tilted when it is inserted to the container engagement mechanism, the level detection mechanism **3102** will detect the tilt, and the medical diagnostic system can inform the user accordingly. The level detection mechanism described above is exemplary, and other level detection mechanisms are contemplated to be within the scope of the present disclosure.

[0106] FIG. **32** is a diagram of the container engagement mechanism **2700** before engaging the caps **3202** of a container. The container can be a reagent container or a working fluid and waste container. In the case of a working fluid and waste container, the two compartment engagement mechanisms **2720** may be separated more than as illustrated. The caps **3202** and the compartment engagement mechanisms **2720** are sized dimensionally to couple with each other. For example, the cap **3202** is sized to contact the wedge-shaped ends of the engagement arms **2726** and to cause the engagement arms **2726** to flex away from the needle cover **2728** and release the needle cover **2728**. Once the needle cover **2728** is no longer grasped by the engagement arms **2726**, the cap **3202** can press against the needle cover **2728** and cause it to slide through or along the hub **2722** and expose the needles **2724**. The needles **2724** puncture the cap **3202** to access the compartment of the container. When the container is fully engaged with the container engagement mechanism **2720**, the engagement arms **2726** are received by the notches in the collar of the container, as described above in connection with FIGS. **27** and **28**, and as shown in FIG. **33**. In various embodiments, the grasping portion of the engagement arms **2726** can grasp the cap **3202** to secure the container within the receptacle, as shown in FIG. **33**.

[0107] Referring again to FIG. **32**, and in accordance with aspects of the present disclosure, one compartment engagement mechanism may be smaller than the other compartment engagement mechanism, such that the container-facing ends of the two compartment engagement mechanisms may not be the same distance from the wall **3210** of the receptacle. Additionally, the caps **3202** of the container may also be different sizes and may extend a different distance from the body of the container. Variations in the sizes and arrangements of the compartment engagement mechanisms **2720** and of the caps **3202** of a container are contemplated to be within the scope of the present disclosure.

[0108] FIG. **34** shows a perspective view of another embodiment of a compartment engagement mechanism, including a hub **3410**, a needle cover **3420**, two engagement arms **3412**, and two needles **3430**. Although the illustrated embodiment includes two needles **3430**, another number of needles can be used in various embodiments, such as fewer than two needles or more than two needles. In various embodiments, multiple needles can be arranged in another manner than as illustrated. As shown in FIG. **34**, the hub **3410** includes guide regions **3414** which receive a portion of the needle cover **3422** and permit the portion of the needle cover **3422** to slide through or along the hub **3410**. In the illustrated embodiment, the engagement arms **3412** are integral with a portion of the hub **3410** and are not removable from the hub **3410**.

[0109] FIG. **35** shows another perspective view of the compartment engagement mechanism, including a container-facing end of the mechanism. As shown in FIG. **35**, the container-facing end of the needle cover **3420** includes apertures **3424**. The apertures **3424** are aligned with the needles **3430** such that the needles **3430** extend through the apertures **3424** when the needle cover **3420** slides through or along the hub **3410**. The needle cover **3420** also includes slots **3426** that receive the engagement arms **3412** of the hub **3410**. FIGS. **34** and **35** are exemplary, and variations of or other embodiments of a compartment engagement mechanism are contemplated to be within the scope of the present disclosure.

[0110] FIG. **36** is a diagram of certain components of the container engagement mechanism of FIGS. **34** and **35**, including a needle cover **3640**, a spring, two hub components **3610**, **3620**, and two needles **3430**. The needles and spring can operate in the same manner described with respect to FIGS. **27-33**.

[0111] In the illustrated embodiment, the hub is formed from two components **3610**, **3620** which combine to form the hub illustrated in FIGS. **34** and **35**. In the illustrated embodiment, the engagement arms **3612** are integral with a main hub component **3610** and are not removable from the hub. The main hub component **3610** also includes apertures **3614** for the needles **3430**. The other hub component **3620** operates as a needle clamp. In the illustrated embodiment, the needles **3430** are first inserted into the apertures **3614** of the main hub component **3610**, and then the other hub component **3620** is fitted onto the needle/hub assembly. In the illustrated embodiment, the other hub component **3620** is structured to clip to the main hub component **3610** and thereby clamp the needles **3630** to the hub. The clip arms **3622** of the other hub component **3620** are oriented perpendicular to the orientation of the engagement arms **3612** of the main hub component **3610**. In various embodiments, the two hub components **3610**, **3620** may engage each other in other ways, such as by friction fit, for example. In various embodiments, the hub may be formed as one unitary component. In various embodiments, the hub can be made from a plastic material.

[0112] Referring again to the main hub component **3610**, the engagement arm **3612** includes an elbow portion **3616** and a grasping portion **3618**. The grasping portion **3618** of the engagement arm **3612** is connected to the elbow portion **3616**. The elbow portion **3616** is semi-flexible so as to flex and permit the grasping portion **3618** to move radially closer to or farther away from the needle cover **3640** of the compartment engagement assembly. The end of the grasping portion **3618** includes a wedge shape pointed towards the needle cover **3640** of the compartment engagement assembly. In various embodiments, the engagement arm **3612** can be made from a plastic material.

[0113] In the illustrated embodiment, the needle cover **3640** includes a cap **3642**, glide posts **3644** attached to the cap **3642**, and slots **3646** that receive the engagement arms **3612** of the hub. In various embodiments, the cap **3642**, the slots **3646**, and the glide posts **3644** can have various shapes, including shapes different from those illustrated in FIG. **36**. In various embodiments, the number of glide posts **3644** can vary, including fewer than four glide posts or more than four glide posts. The glide posts **3644** slide through or along the hub. As shown in FIGS. **34** and **35**, each glide post **3644** contacts both of the two hub components **3610**, **3620**, such that the two hub components **3610**, **3620** cooperate to provide guide regions for the glide posts **3644** of the needle cover **3640**. In various embodiments, the number of slots **3646** of the needle cover **3640** can correspond to the number of engagement arms **3612**. In various embodiments, the needle cover **3640** can be made from a plastic material.

[0114] FIG. **37** shows another perspective of various components shown in FIG. **36**. In various embodiments, the hub **3610** can include a spring alignment structure **3702** that operates to align the spring **3704** and also serves as a stop for the retracting needle cover **3640**. The shape of the illustrated structure **3702** is exemplary, and other variations are contemplated to be within the scope of the present disclosure. In various embodiments, the needle cover **3640** also includes a spring alignment structure (not shown), which can serve as a dead stop against the spring alignment structure **3702** of the hub **3610**. In various embodiments, the spring alignment structure of the needle cover **3640**, if any, can be the same shape as the spring alignment structure **3702** of the hub **3610** or can be a different shape. The illustrated and described embodiments are exemplary, and variations are contemplated to be within the scope of the present disclosure.

[0115] The embodiments disclosed herein are examples of the disclosure and may be embodied in various forms. For instance, although certain embodiments herein are described as separate embodiments, each of the embodiments herein may be combined with one or more of the other embodiments herein. Specific structural and functional details disclosed herein are not to be interpreted as limiting, but as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed structure. Like reference numerals may refer to similar or identical elements throughout the description of the figures.

[0116] The phrases “in an embodiment,” “in embodiments,” “in various embodiments,” “in some embodiments,” “in various embodiments,” or “in other embodiments” may each refer to one or more of the same or different embodiments in accordance with the present disclosure. A phrase in the form “A or B” means “(A), (B), or (A and B).” A phrase in the form “at least one of A, B, or C” means “(A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).”

[0117] It should be understood that the foregoing description is only illustrative of the present disclosure. Various alternatives and modifications can be devised by those skilled in the art without departing from the disclosure. Accordingly, the present disclosure is intended to embrace all such alternatives, modifications and variances. The embodiments described with reference to the attached drawing figures are presented only to demonstrate certain examples of the disclosure. Other elements, steps, methods, and techniques that are insubstantially different from those described above and/or in the appended claims are also intended to be within the scope of the disclosure.

[0118] The systems described herein may also utilize one or more controllers to receive various information and transform the received information to generate an output. The controller may include any type of computing device, computational circuit, or any type of processor or processing circuit capable of executing a series of instructions that are stored in a memory. The controller may include multiple processors and/or multicore central processing units (CPUs) and may include any type of processor, such as a microprocessor, digital signal processor, microcontroller, programmable logic device (PLD), field programmable gate array (FPGA), or the like. The controller may be located within a device or system at an end-user location, may be located within a device or system at a manufacturer or servicer location, or may be a cloud computing processor located at a cloud computing provider. The controller may also include a memory to store data and/or instructions that, when executed by the one or more processors, causes the one or more processors to perform one or more methods and/or algorithms.

[0119] It should be understood that the foregoing description is only illustrative of the present disclosure. Various alternatives and modifications can be devised by those skilled in the art without departing from the disclosure. Accordingly, the present disclosure is intended to embrace all such alternatives, modifications and variances. The embodiments described with reference to the attached drawing figures are presented only to demonstrate certain examples of the disclosure. Other elements, steps, methods, and techniques that are insubstantially different from those described above and/or in the appended claims are also intended to be within the scope of the disclosure.

Claims

1. A method for engaging, by a system, a container storing a fluid to be utilized in a diagnostic process, the system comprising: a receptacle; a hub fixed in a wall of the receptacle; at least one needle secured to the hub and protruding from the hub; a needle cover slideably coupled with the hub, the needle cover comprising at least one aperture corresponding to the at least one needle; a spring positioned between the hub and the needle cover; and an engagement arm having a base portion secured to the hub and a grasping portion, the method comprising: pushing, by the spring in a first compressed state, the needle cover against the grasping portion of the engagement arm such that the grasping portion grasps the needle cover in the first compressed state of the spring; receiving, in the receptacle, the container storing the fluid; and compressing the spring, from the first compressed state to a second compressed state, by the container pressing against the needle cover in the receptacle, wherein, while compressing the spring from the first compressed state to the second compressed state, the at least one needle protrudes through the at least one aperture of the needle cover and punctures the container.

2. The method of claim 1, wherein the needle cover comprises a cap defining the at least one aperture corresponding to the at least one needle.

3. The method of claim 2, wherein in the first compressed state of the spring, the at least one needle is covered by the cap such that the at least one needle extends from the hub and terminates prior to the at least one aperture of cap.

4. The method of claim 2, wherein the hub defines a plurality of guide slots, wherein the needle cover further comprises a plurality of glide posts extending from the cap, and wherein individual glide posts of the plurality of glide posts are positioned at least partially within individual guide slots of the plurality of guide slots of the hub.

5. The method of claim 4, wherein, while compressing the spring from the first compressed state to the second compressed state, the plurality of glide posts slide along the plurality of guide slots.

6. The method of claim 1, wherein the spring contacts the hub and the needle cover in the second compressed state.

7. The method of claim 1, wherein, while compressing the spring from the first compressed state to the second compressed state, the spring maintains contact with the hub and the needle cover.

8. The method of claim 1, further comprising: releasing the needle cover by the grasping portion of the engagement arm as the spring compresses from the first compressed state to the second compressed state.

9. The method of claim 1, wherein the hub further comprises a spring alignment structure configured to align the spring against the hub, and wherein the needle cover comprises a spring alignment structure configured to align the spring against the needle cover.

10. The method of claim 9, wherein in the second compressed state of the spring, the spring alignment structure of the hub contacts the spring alignment structure of the needle cover.

11. The method of claim 1, further comprising: decompressing the spring from the second compressed state to the first compressed state by the container being moved away from the needle cover.

12. The method of claim 11, further comprising: re-grasping the needle cover by the grasping portion of the engagement arm as the spring decompresses from the second compressed state to the first compressed state.

13. The method of claim 11, wherein the hub defines a plurality of guide slots, wherein the needle cover comprises: a cap defining the at least one aperture corresponding to the at least one needle, and a plurality of glide posts extending from the cap, and wherein individual glide posts of the plurality of glide posts are positioned at least partially within individual guide slots of the plurality of guide slots of the hub.

14. The method of claim 13, wherein, while decompressing the spring from the second compressed state to the first compressed state, the plurality of glide posts of the needle cover slide along the plurality of guide slots of the hub.